



Central Coast Regional Water Quality Control Board

July 19, 2022

Sent Via Electronic Mail

Elizabeth R. Clark
Environmental Division Chief, Director of Public Works
USAG Fort Hunter Liggett
Building 238 California Ave.
Fort Hunter Liggett, CA 93928-7000
Email: elizabeth.r.clark14.civ@army.mil

Dear Elizabeth Clark,

**UNITED STATES ARMY GARRISON FORT HUNTER LIGGETT WASTEWATER
TREATMENT PLANT, JOLON, MONTEREY COUNTY - TRANSMITTAL OF REVISED
MONITORING AND REPORTING PROGRAM ORDER NO. R3-2022-0025**

Central Coast Regional Water Quality Control Board staff revised Monitoring and Reporting Program R3-2022-0025 for the United States Army Garrison (USAG) Fort Hunter Liggett Wastewater Treatment Plant's enrollment into State Water Resources Control Board's General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems, Order WQ 2014-0153-DWQ. A copy of the revised monitoring and reporting program is attached.

Please be aware that this revised monitoring and reporting program alters certain sampling and reporting requirements and USAG is responsible for ensuring that those requirements are met.

Please contact Mark Lemus at (805) 549-3703 Mark.Lemus@Waterboards.ca.gov or Jennifer Epp (805) 594-6181 Jennifer.Epp@Waterboards.ca.gov with any questions you may have concerning this matter.

Sincerely,

Jennifer Epp
Digitally signed by Jennifer Epp
Date: 2022.07.19 12:17:31 -07'00'
Water Boards

for, Matthew T. Keeling,
Executive Officer

JANE GRAY, CHAIR | MATTHEW T. KEELING, EXECUTIVE OFFICER

Attachment: Revised Monitoring and Reporting Program Order No. R3-2022-0025

cc:

Michael Moeller, Michael.b.moeller.civ@army.mil
Todd Stanley, Todd.Stanley@Waterboards.ca.gov
Jennifer Epp, Jennifer.Epp@Waterboards.ca.gov
WDR Program, RB3-WDR@Waterboards.ca.gov

ECM/CIWQS = CW-225534

GeoTracker No. = GT-WDR100037150

Rev 3/11/2022

ECM Subject Name = USAG Fort Hunter Liggett Revised MRP Transmittal Letter
R:\RB3\Shared\WDR\WDR Facilities\Monterey Co\Fort Hunter Liggett\2022\General
Order Enrollment\0153 NOA and MRP\Revised
MRP\FHL_0153_Revised_MRP_&Transmittal_Letter.docx

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

REVISED MONITORING AND REPORTING PROGRAM NO. R3-2022-0025

for

**United States Army Garrison Fort Hunter Liggett
MONTEREY COUNTY**

This Monitoring and Reporting Program (MRP) describes requirements for monitoring a wastewater treatment system for United States Army Garrison (USAG) Fort Hunter Liggett.

USAG owns and operates the wastewater system that is subject to the notice of applicability dated March 21, 2022, for coverage in *Order WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems* (General Permit).

1. SAMPLING AND ANALYSIS

All samples must be representative of the volume and nature of the discharge or matrix of materials sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample shall be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom the samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) shall be approved by the Central Coast Regional Water Quality Control Board (Central Coast Water Board). Unless otherwise specified below, sampling shall be performed as shown in Table 1.

Table 1. Sampling and Monitoring Schedule

Monitoring Period	Sample Collection Months
Annually	October
Semiannually	April and October
Quarterly	January, April, July, and October
Monthly	Each month of the year

Laboratories analyzing water samples (wastewater and groundwater) must be accredited by the State Water Resources Control Board (State Water Board) Environmental Laboratory Accreditation program, in accordance with California Water Code Section 13176, and must include quality assurance/quality control data with their reports. Field test instruments (such as those to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a State Water Board Environmental Laboratory Accreditation Program certified laboratory, or:

1. The user is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are maintained and available for at least three years.

2. WATER SUPPLY MONITORING

Representative samples of the water supply must be collected and analyzed as shown in Table 2 from the production wells prior to any disinfection.

In lieu of the required water supply sampling, USAG may request approval to submit a Well Identification Number and the reporting year's consumer confidence report (annual water quality report or drinking water quality report), as required by the State Water Board Division of Drinking Water and/or county. USAG must report the results of any constituent monitored more frequently than is required by the monitoring program. USAG must also report detectable concentrations (above the reporting limit) for any other constituent that has a published maximum contaminant level (MCL).

Table 2. Water Supply Monitoring

Parameter	Units	Type of Sample	Sampling Frequency ²	Reporting Frequency ²
TDS	mg/L ¹	Grab	Once every three years	Once every three years
Nitrate (as N)	mg/L	Grab	Once every three years	Once every three years
Chloride	mg/L	Grab	Once every three years	Once every three years
Sodium	mg/L	Grab	Once every three years	Once every three years
Sulfate	mg/L	Grab	Once every three years	Once every three years
Carbonate	mg/L	Grab	Once every three years	Once every three years
Bicarbonate	mg/L	Grab	Once every three years	Once every three years
Calcium	mg/L	Grab	Once every three years	Once every three years
Magnesium	mg/L	Grab	Once every three years	Once every three years

¹ mg/L denotes milligram per liter

² Once every three years or at a frequency to coincide with potable drinking water well sampling and reporting requirements.

3. MONITORING LOCATIONS

Monitoring location information, including nomenclature that shall be used when reporting data to the GeoTracker database (described below), is shown in Table 3.

Table 3. Monitoring Locations

Sample Title	GeoTracker Field Point Code	Sample Description
Influent Sample – 1	IS-1	Representative grab sample from influent distribution box after screening before lagoons
Effluent Sample – 1	ES-1	Representative grab sample from online lagoon effluent weir before disposal
Monitoring Well - 1	MW-1	Representative grab sample from monitoring well 1
Monitoring Well - 2	MW-2	Representative grab sample from monitoring well 2
Monitoring Well - 3	MW-3	Representative grab sample from monitoring well 3

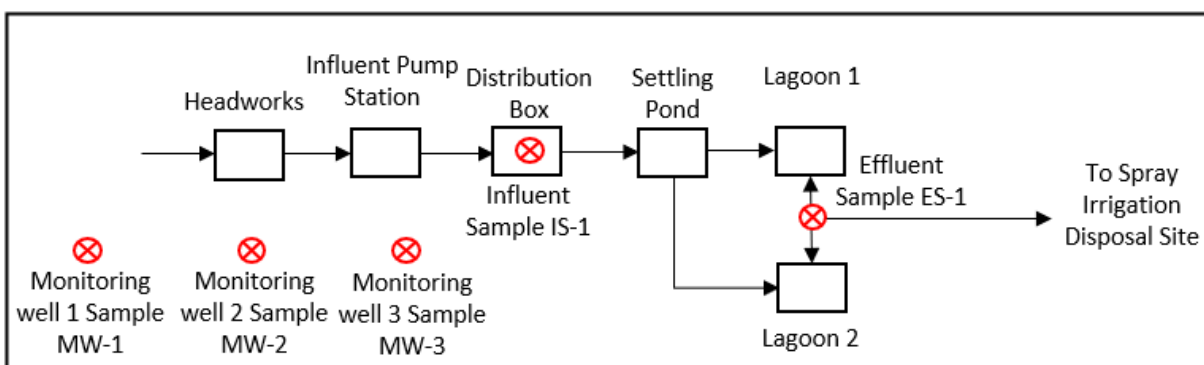


Figure 1. Process Flow Diagram with Sampling Locations

4. POND SYSTEM MONITORING

A. Flow Monitoring

USAG must monitor and report flow, as described in Table 4, with each monitoring report. Flow is monitored at via Parshall flume/ultrasonic flow meter at the headworks.

Table 4. Flow Monitoring

Parameter	Units	Sample Type	Frequency
Daily Influent Flow	gpd ¹	Metered	Daily
Maximum Daily Influent Flow	gpd	Metered	Monthly
Mean Daily Influent Flow	gpd	Calculated	Monthly

¹ gpd denotes gallons per day

Influent and Effluent Constituent Monitoring

Representative samples of USAG's influent (raw wastewater) into the wastewater system and effluent (treated wastewater) discharged to land must be collected and analyzed as described in Table 5. Effluent samples shall only be collected in months and at times when spray disposal is occurring. All samples must be grab.

Table 5. Influent and Effluent Monitoring

Parameter/ Constituent	Units	INFLUENT (IS-1) Sampling Frequency	EFFLUENT (ES-1) Sampling Frequency
Biochemical Oxygen Demand, 5-Day	mg/L ¹	Monthly	Monthly
Total Suspended Solids	mg/L	Monthly	Monthly
pH	pH units	Monthly	Monthly
Total Nitrogen	mg/L	Quarterly	Monthly
Nitrate (as N)	mg/L	Quarterly	Monthly
Total Kjeldahl Nitrogen (as N)	mg/L	Quarterly	Monthly
Total Dissolved Solids	mg/L	Not Required	Semi-annually
Chloride	mg/L	Not Required	Semi-annually
Sodium	mg/L	Not Required	Semi-annually
Sulfate	mg/L	Not Required	Semi-annually

¹ mg/L denotes milligram per liter

Wastewater Pond Monitoring

All wastewater ponds and storage ponds must be monitored consistent with Table 6. USAG must inspect the ponds weekly and notations must be made in a bound logbook and include “observations” of the items in Table 6. Any problems must be promptly corrected and recorded. The bound logbook of these observations must be maintained onsite and made available to the Central Coast Water Board upon inspection. Only violations and mitigation measures must be included in the subsequent monitoring report.

Table 6. Wastewater Pond Monitoring

Parameter	Units	Sample Type	Sample Frequency
Dissolved Oxygen	mg/L ¹	Grab ²	Weekly
pH	Std. units	Grab ²	Weekly
Evaporation	Inches	Weather station ³	Daily
Freeboard	0.1 feet	Measurement	Daily
Odors	--	Observation	Weekly
Berm Condition	--	Observation	Weekly
Plant Growth	--	Observation	Weekly
Vectors or Animals Burrowing	--	Observation	Weekly
Disposal Area Condition	--	Observation	Weekly

¹ mg/L denotes milligrams per liter

² Grab samples can be collected via handheld probe, see section 1 of this MRP

³ Weather station may be site-specific station, onsite evaporation pan, or nearby governmental weather reporting station

5. LAND APPLICATION AREA MONITORING

USAG shall monitor the Spray Irrigation Disposal Site when wastewater irrigation is applied. If wastewater irrigation is not applied during a reporting period, the monitoring report shall so state. Precipitation, windspeed and direction shall be recorded regardless if wastewater irrigation is currently being applied. Spray Irrigation Disposal Site monitoring shall include the following:

Table 7. Land Application Area Monitoring

Parameter	Units	Sample Type	Sample Frequency
Daily Effluent Flow	gpd ¹	Metered/Estimated ²	Semiannually
Total Flow	gallons	Calculated	Monthly
Acreage Applied ³	Acres	Calculated	Monthly
Application Rate	gal/acres/mo	Calculated	Monthly
Local Rainfall	Inches	Weather Station ⁴	Each Precipitation Event
Wind Speed & Direction	mph ⁵	Weather Station	Daily
Irrigation Equipment	--	Observations ⁶	Weekly
Nuisance Odor Condition	--	Observations	Weekly
Saturated Soil/Ponding	--	Observations	Weekly
Plant Growth	--	Observations	Weekly
Vectors or Animals Burrowing	--	Observations	Weekly
Evidence of Soil Erosion	--	Observations	Weekly

¹ gpd denotes gallons per day

² Effluent flow can be determined via meter reading, or estimated with pump time meter calculations.

³ Acreage applied denotes the acreage to which wastewater is applied

⁴ Weather station may be site-specific station, onsite gauge, or nearby governmental weather reporting station.

⁵ mph denotes miles per hour

⁶ Observations must be maintained onsite in a bound logbook and made available to the Central Coast Water Board upon inspection. Any problems must be promptly corrected and recorded.

6. SOLIDS DISPOSAL MONITORING

USAG must report the handling and disposal of all solids (e.g., screenings, grit, sludge, biosolids, grease, etc.) generated at the wastewater treatment system. Records must include the name/contact information for the hauling company, the type and amount of waste transported, the date removed from the wastewater system, the disposal facility name and address, and copies of analytical data required by the entity accepting the waste. These records must be submitted as part of the subsequent monitoring report.

7. GROUNDWATER MONITORING

Prior to any sampling, depth to groundwater must be measured and groundwater elevations must be calculated. The monitoring wells must be purged of at least three well volumes and until the measurements of the following parameters have stabilized (i.e., are reproducible with 10 percent): pH, temperature, dissolved oxygen, electrical

conductivity, and turbidity. No-purge, low flow, or other sampling techniques are acceptable if they are described in the Central Coast Water Board approved sampling and analysis plan. Once the groundwater level in each of the wells has recovered sufficiently to ensure the collection of a representative groundwater samples, a qualified individual (e.g., consultant, technician, etc.) trained in using proper sampling methods shall recover samples using approved USEPA methods. All data must be recorded and submitted in monitoring well field sheets.

USAG shall monitor all three groundwater monitoring wells surrounding the treatment lagoons as follows:

Table 8. Groundwater Monitoring

Parameter	Units	Sample Type	Sample Frequency
Depth to Groundwater	0.01 feet	Measurement	Quarterly
Groundwater Elevation	0.01 feet	Calculated	Quarterly
Gradient	0.01 feet	Calculated	Quarterly
Gradient Direction	degrees	Calculated	Quarterly
pH	Std. Units	Grab	Quarterly
Total dissolved Solids	mg/L ¹	Grab	Quarterly
Nitrate as (N)	mg/L	Grab	Quarterly
Sodium	mg/L	Grab	Quarterly
Chloride	mg/L	Grab	Quarterly
Sulfate	mg/L	Grab	Quarterly
Boron	mg/L	Grab	Quarterly

¹ mg/L denotes milligrams per liter

8. INFLOW AND INFILTRATION (I/I) REDUCTION

USAG must report ongoing efforts to reduce I/I annually. Information should include at minimum the following:

- Continued monitoring and reporting on precipitation events and their correlation to increased flows received at the wastewater treatment plant.
- Ongoing investigations and identification of any other sources contributing to I/I.
- Manhole repair/replacement activities conducted during the previous year.
- Sewer pipe repair/replacement work conducted during the previous year
- A working plan for the upcoming years activities.
- Maps of defects addressed in the previous year and those that the USAG will be addressing the following year.

9. REPORTING

The semiannual monitoring reports must be submitted in accordance with Table 9.

Table 9. Semiannual Monitoring Reports

Report	Monitoring Period	Report Due Date
First Semiannual Report	January 1 to June 30	August 1
Second Semiannual Report	July 1 to December 31	February 1

At a minimum, the semiannual reports must include:

- **Facility Information and Description** -- Briefly describe your facility, treatment process, and disposal process with references to figures.
- **Data Tables** -- Tabular summaries of all monitoring data (water supply, influent, effluent, and groundwater) compared to effluent limits, as appropriate. Submit the results of any pollutant or parameter monitored more frequently than is required by this monitoring program.
- **Compliance and Performance Discussion** -- Disclosure of any noncompliance (violations) with of the notice of applicability and/or General Permit requirements, including a description, it's cause, the exact date(s) and time(s), and the steps taken or planned to reduce, eliminate and prevent recurrence of the noncompliance.
An evaluation (graphs recommended) of the treatment facilities' performance, including percent removal of the main pollutants (BOD₅, TSS, Total Nitrogen, Total Coliforms) over time, nuisance conditions, system problems, etc.
A discussion of any data gaps and potential deficiencies in the monitoring system or reporting program.
- **Disposal Area Inspection** -- Report the results of the disposal area inspections, as described in Section 6.
- **Solids Disposal** -- If solids are disposed of offsite, comply with the reporting in Section 7.
- **Operator Information** -- List the name and contact information for the wastewater operators responsible for operation, maintenance, and/or system monitoring.
- **Flow Evaluation** -- A summary in tabular format of the maximum and average flows for each month and comparison to permitted flows.
- **Figures** -- A wastewater treatment process flow diagram with a label for the monitoring locations and a scaled facility map showing the treatment system, discharge points, and wells (supply and groundwater, if present).
- **Laboratory Sheets** -- Copies of laboratory analytical report(s) and chain of custody form(s).
- **Inflow and Infiltration** -- Annually, (in the first semi-annual report), report ongoing efforts and progress to reduce I/I annually as described above.

A. Electronic Submittal

All monitoring reports must be provided electronically in a searchable PDF format, with the Central Coast Water Board's current transmittal sheet found at the link below as the cover page. The transmittal sheet must be signed.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permittting/docs/transmittal_sheet.pdf

USAG must submit all reports/documents and laboratory analytical data (influent, effluent, groundwater data) to the State Water Board's GeoTracker^{1,2} database, using specific global identification number **WDR100037150**, consistent with applicable Electronic Submittal of Information (ESI) requirements under a Wastewater System-specific global identification number over the internet at:

http://www.waterboards.ca.gov/ust/electronic_submittal/index.shtml.

Table 10 summarizes the GeoTracker electronic reporting requirements. For general questions, please contact the GeoTracker Help Desk: Geotracker@waterboards.ca.gov.

Table 10. GeoTracker Electronic Submittal Information (ESI) Data Requirements

Electronic Submittal	Description of Action	Action	Frequency
Reports and Documents	Complete copy of all documents including monitoring reports (in searchable PDF format) and any other associated documents related to the Wastewater System.	Upload directly to GeoTracker all monitoring reports (in searchable PDF format) and any other associated documents.	On or before the due dates required by this General Permit and for other documents when required by the Central Coast Water Board.
Laboratory Data	All analytical data (including geochemical data) in electronic deliverable format (EDF). This includes all water, soil, and vapor samples collected when monitoring a discharge.	Upload, or direct your State Certified Laboratory staff to upload, all laboratory data directly to GeoTracker.	On or before the due date of the required monitoring report

¹ Information for first-time GeoTracker users is available at:

https://www.waterboards.ca.gov/ust/electronic_submittal/docs/beginnerguid2.pdf

² Additional information available at:

<https://geotracker.waterboards.ca.gov/>

Electronic Submittal	Description of Action	Action	Frequency
Depth to Groundwater	Monitoring wells must have the depth-to-water information reported. Report data only for wells defined as permanent sampling points.	Upload depth-to-water information to the GeoTracker GEO_WELL file.	On or before the due date of the required monitoring report
Boring Logs and Well Screen Intervals	Boring logs must be prepared by a registered professional and submitted in PDF format separately (not only as attachments to reports)	Upload boring logs (in searchable PDF format) to GeoTracker whenever a new boring is drilled.	Every time a new boring is drilled.
Location data (Geo XY)	Name, classify, and identify the location (latitude and longitude) of all sampling points. Monitoring wells must be surveyed, influent and effluent sample locations must be identified on the GeoTracker mapping tool under "non-surveyed data."	Upload the location data (surveyed and non-surveyed) to the GeoTracker Geo_XY file.	These data points are required prior to laboratory data uploads. Must be added every time a permanent monitoring point is established.
Elevation Data (Geo Z)	Mark the elevation at the top of groundwater well casings for all permanent groundwater wells. These points are required prior to depth-to-water data uploads.	Upload the survey data to the GeoTracker GEO_Z File.	One-time, for all groundwater monitoring wells.
Geo Map	Site layout, map of facilities, wastewater treatment system, and disposal area(s).	Upload the Site layout PDF to the GeoTracker site plan file.	One time, or when the facility is modified.

10. TECHNICAL REPORTS

USAG must submit to GeoTracker the three reports described in General Permit *Section E. 1. Technical Report Preparation Requirements* by the dates included in Table 11. If this information is already contained in an Operation and Maintenance Manual or similar document, that may be submitted in compliance with this requirement.

Table 11. Technical Report Submittal Due Dates

Report	Report Due Date
Spill Prevention & Emergency Response Plan	90 days from date of General Permit enrollment ¹
Sampling and Analysis Plan	90 days from date of General Permit enrollment ¹
Sludge Management Plan	90 days from date of General Permit enrollment ¹
I/I Reduction Work Plan	6 months from date of General Permit enrollment ¹
Irrigation Spray Disposal Repair Work Plan	6 months from date of General Permit enrollment ¹

¹. The date of General Permit enrollment is March 21, 2022

In addition to the technical reports required in section E.1. of the General Permit, USAG must develop the following technical reports:

A. Infiltration and Inflow (I/I) Reduction Work Plan

USAG will develop a onetime technical work plan report to address their current I/I issues that result in increased flow and potential violations. This report should provide information demonstrating that the USAG is producing measurable and successful efforts to reduce I/I. The work plan, at a minimum, must contain the following:

1. A narrative description of the work performed to date to reduce I/I.
2. A current map of sanitary sewer collection system defects in need of repair.
3. A list of repairs/defects to be addressed
4. A timeline to repair/install a new influent flow meter.
5. A costs analysis projection for work to be completed.
6. A proposed timeline to repair/replace sanitary sewer collection system defects.

B. Spray Irrigation Disposal Site Repair Work Plan

USAG will develop a onetime technical work plan report to address their current spray irrigation disposal system broken pipes and sprinklers. This report should provide information demonstrating that USAG is producing measurable and successful efforts to repair and replace broken irrigation pipelines and sprinkler heads to ensure that the facility is land applying treated wastewater effluent as originally designed and in compliance with appropriate land application requirements of the General Permit. The work plan, at a minimum, must contain the following:

1. A narrative description of the current status of the spray irrigation disposal system.
2. A map of the current irrigation piping layout with noted defects.
3. A list of piping and sprinkler defects.
4. Establishment of a preventative maintenance program to provide ongoing repair work and planned replacements during the remaining useful life of the disposal system.
5. A costs analysis projection for work to be completed.
6. A proposed timeline to repair/replace irrigation piping and sprinkler defects.

11. LEGAL REQUIREMENTS

All technical and monitoring reports submitted pursuant to this Monitoring and Reporting Program are required pursuant to section 13267 of the California Water Code. Failure to submit reports in accordance with schedules established by this General Permit or attachments to this General Permit, or failure to submit a report of sufficient technical quality to be acceptable to the Central Coast Water Board, may subject USAG to enforcement action pursuant to section 13268 of the California Water Code.

Water Code section 13267 states, in part:

“In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports.”

Water Code section 13268 states, in part:

“(a) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).

(b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a

violation of subdivision (a) in an amount which shall not exceed one thousand dollars (\$1,000) for each day in which the violation occurs.

The burden and cost of preparing the reports is reasonable and consistent with the interest of the state in maintaining water quality. The reports are necessary to ensure that USAG complies with the notice of applicability and General Permit. Pursuant to Water Code section 13267, USAG must implement this MRP and must submit the monitoring reports described herein.

USAG must implement the above monitoring program as of the date of this MRP. The Central Coast Water Board may rescind or modify the MRP at any time.

Ordered by:

Jennifer Epp  Digitally signed by Jennifer Epp
Date: 2022.07.19 12:17:51 -07'00'
Water Boards

for Matthew T. Keeling
Executive Officer